

3 The photograph shows a sunflower plant, *Helianthus annuus*.



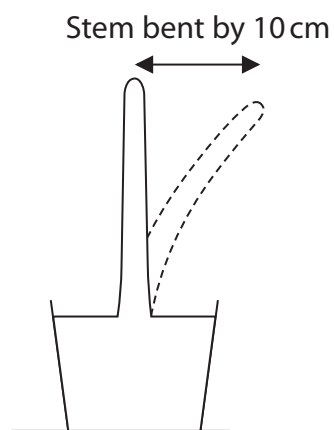
(Source: © Richard Griffin/Alamy Stock Photo)

A student investigated the effect that wind had on the bending of plants.

Thirty sunflower plants were grown in individual pots until they were 60 cm tall.

The leaves and flowers were removed from the sunflower plants.

For one group of 15 plants, the top of each stem was bent by 10 cm and then released as shown in the diagram.



The bending of the stems was repeated once a day for eight weeks.

A control group of 15 plants were kept in the same conditions but without the stems being bent.

The diameter of the stem at the base of each plant was measured after eight weeks.

The results of this investigation are:

Experimental group stem diameters (mm)

17.34	18.16	17.61	18.14	17.64	18.14	17.95	17.78
17.69	17.84	17.99	17.95	17.89	18.16	18.22	

Control group stem diameters (mm)

17.15	17.52	17.57	17.03	17.26	17.31	17.09	17.38
17.41	17.21	17.34	17.16	17.15	17.32	17.60	

(a) State a suitable null hypothesis for this investigation.

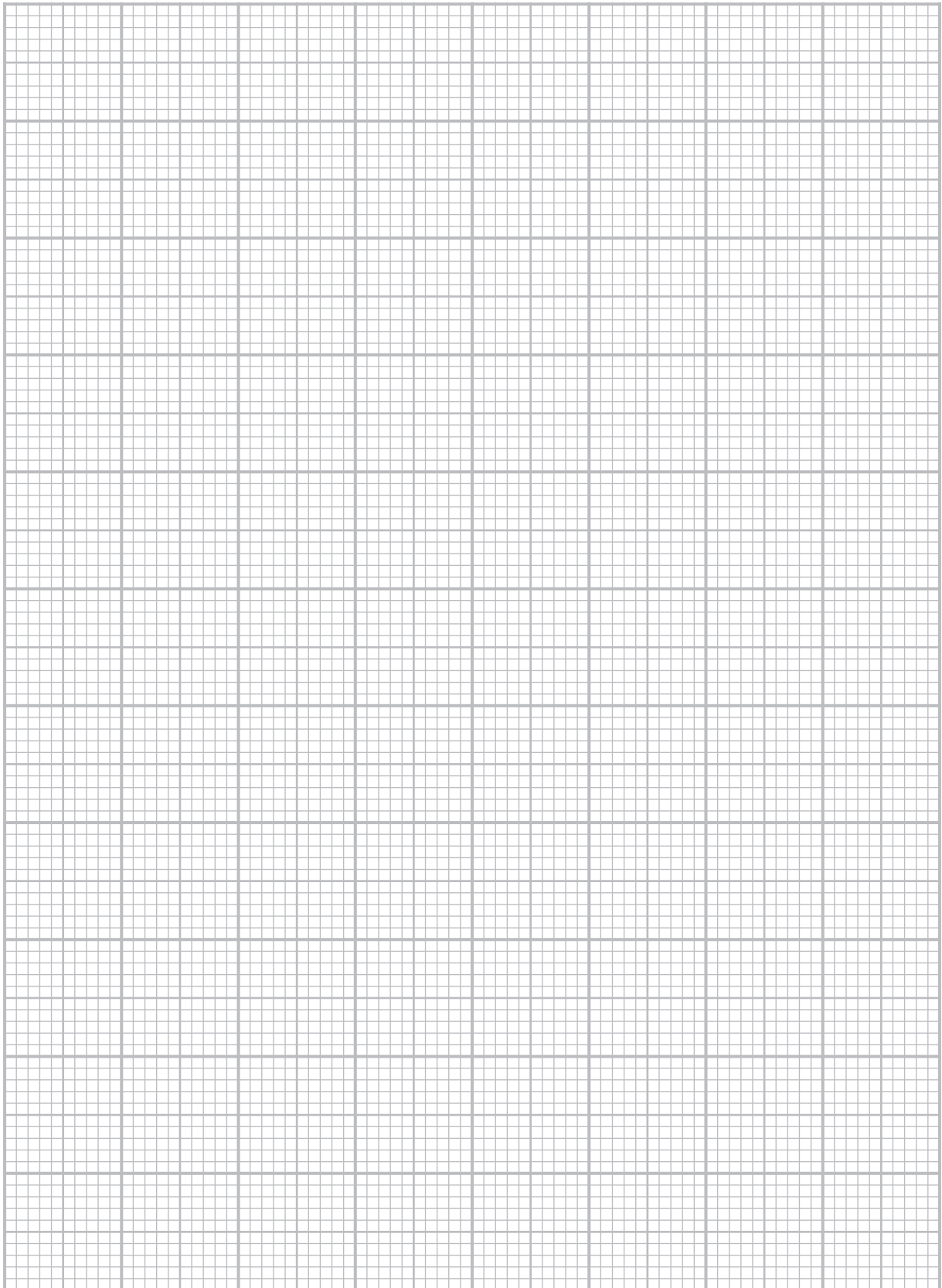
(1)

(b) Draw a suitable table to display the **data** and your calculated **means** for these two groups of plants.

(3)

- (c) Plot a suitable graph to show the mean diameter of the stems for each group of plants. Include an indication of the variability of these data.

(3)



(d) The student analysed this data with a t test using the formula:

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

where:

$\bar{x}$ is the mean value for each group of plants

n is the number of samples for each group of plants

$(S_A)^2 = 0.03$ and $(S_B)^2 = 0.06$

(i) Calculate the value of t .

(2)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_A - 1) + (n_B - 1)$

Degrees of freedom	Level of significance (p)	
	0.05	0.01
15	2.13	2.95
16	2.12	2.92
17	2.11	2.90
18	2.10	2.88
19	2.09	2.86
20	2.09	2.84
21	2.08	2.83
22	2.07	2.82
23	2.07	2.81
24	2.06	2.80
25	2.06	2.79
26	2.06	2.78
27	2.05	2.77
28	2.05	2.76
29	2.04	2.76
30	2.04	2.75

Deduce the conclusions that can be drawn from this investigation.
Use your graph and the information in the table to support your answer.

(2)

(e) Explain why it may **not** be reasonable to draw valid conclusions from the results of this investigation.

(2)

(Total for Question 3 = 13 marks)